



# RAMCO INSTITUTE OF TECHNOLOGY

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## Department of Computer Science and Engineering

Academic Year 2023 - 2024 (Odd Semester)

Degree, Semester & Branch: B.E- V Semester-CSE

Course Code & Title: CB3491 - Cryptography and Cyber Security

Name of the Faculty member (s): Mrs.K.Jeyakarthika, AP/CSE

Date of Implementation: 01.08.2023

### Active Learning method- Description

**Unit / Topic:** Unit I / Classical Encryption Techniques

**Course Outcome:** CO1

**Topic Learning Outcome:** TLO3

**Activity Chosen:** One Minute Paper

**Justification:** In unit I, The Play fair Cipher encryption technique can be used to encrypt or encode a message. It operates exactly like typical encryption. The only difference is that it encrypts a digraph, or a pair of two letters, instead of a single letter. The Play fair cipher uses a  $5 \times 5$  table containing a key word or phrase.

**Time Allotted for the Activity:** 5 Minutes

### Details of the Implementation:

- Faculty explained the topic Play fair in class.
- The time allotted for the topic is 10 minutes
- Faculty give some exercise, students were asked to write in a paper within 5 minutes as shown in Fig 1.
- The papers were collected and corrected. Some student's written wrong answer, that will be discussed in the next day class.
- This activity shows whether students understand the concepts, and to analyse whether they are actively participating in the lesson.

### CO – PO / PSO mapping:

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO12 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|
| CO1 | 3   | 2   | 1   | 2   | 2   | -   | -   | 1   | 1   | 1    |

**PO / PSO mapped:**

| Innovative Practice | PO1                                                                                                                   | PO2                                                                    | PO3                                                                                 | PO4                                                                | PO5                                                                                                     | PO8                                                             | PO9                                                                                              | PO12                                                                                                        |
|---------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
|                     | 3                                                                                                                     | 2                                                                      | 1                                                                                   | 2                                                                  | 2                                                                                                       | 1                                                               | 1                                                                                                | 1                                                                                                           |
|                     | Basic Computer Science Engineering and mathematics Knowledge for performing classical encryption is highly correlated | Analyzing various classical encryption algorithms is highly correlated | Developing systems with basic classical encryption mechanisms are highly correlated | Investigate various aspects of cryptanalysis is highly correlated. | Use of virtual environment and crypt tool for performing encryption and decryption is highly correlated | Students will complete the assignments/ tasks in an ethical way | The students will work as an individual and team for how to use classical encryption Algorithms. | Updating the classical encryption and decryption algorithms to the current trends are moderately correlated |

**Images / Screenshot of the practice:**

953621164009

Cyber Security

Day fair

Plaintext: PRESERVE

Key: AGRICULTURE

|   |   |   |   |   |
|---|---|---|---|---|
| A | G | R | I | C |
| U | L | T | E | B |
| D | F | H | J | K |
| M | N | O | P | Q |
| V | W | X | Y | Z |

PR = RT  
ES = BQ  
ER = IT  
VE = YU

Plaintext: PRESERVE

Ciphertext: RTBQITYU

Plaintext: PRESERVE

Key: AGRICULTURE

|   |   |   |   |   |
|---|---|---|---|---|
| A | G | R | I | C |
| U | L | T | E | B |
| D | F | H | K | M |
| N | O | P | Q | S |
| V | W | X | Y | Z |

PR - XT  
ES - BQ  
ER - TI Q TJ  
VE - YU

playfair cipher:

\* Playfair cipher is 5x5 matrix key is filled in the matrix with no repeated letters.

\* Given plaintext split two words if similar

**Fig 1: One Minute Paper**

### **Reflective Critique:**

#### **❖ Feedback of practice from students and other stakeholders:**

- The students felt that they can able to understand the topic clearly after conducting this activity.
- Also, they can able to practice more problems on classical Encryption Techniques.

#### **❖ Benefit of the practice:**

- Students were actively participated in this activity.
- As a faculty, I could identify the level of understanding swing components.

#### **❖ Challenges faced in implementation:**

- Few students made a mistake while doing exercise problem for play fair cipher.
- The time allotted is 5 minutes, but some students took 10 minutes to complete this task.

### **References:**

- [https://www.ritrjpm.ac.in/images/computer-science/2022-2023/4.VA\\_One-MinutePaper.pdf](https://www.ritrjpm.ac.in/images/computer-science/2022-2023/4.VA_One-MinutePaper.pdf)
- <https://www.ritrjpm.ac.in/images/computer-science/2022-2023/OMP-RV-IIIYR-CSE-CS8601-MC-UII.pdf>

Signature of Faculty Member

HOD